

# Describing Meaning – Predicates and Sets: Exercise Sheet

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## Exercise 1: Predication as membership

For each sentence: (i) treat the predicate as a set (e.g.  $\llbracket \text{cat} \rrbracket = \{x : x \text{ is a cat}\}$ ), (ii) give a set-theoretic representation using  $\in$ , (iii) give an equivalent predicate-notation representation (e.g.  $\text{cat}(\text{kiki})$ ).

- a. Kiki is a dog.
  - b. Rory is tall.
  - c. Mina sleeps.
  - d. Aoife is smart.
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## Exercise 2: Set-builder notation and cardinality

- a. Write each set both extensionally (by listing members) and intensionally (set-builder notation):
  - i. The set of integers  $n$  such that  $-3 \leq n \leq 3$ .
  - ii. The set of even numbers  $\leq 10$ .
  - iii. The set of vowels in the English alphabet.
- b. Compute the cardinality of each set:
  - i.  $\{2, 3, \{4, 5, 6\}\}$
  - ii.  $\emptyset$
  - iii.  $\{\emptyset\}$
  - iv.  $\{\emptyset, \{a, b\}, a\}$

$$v. \{\{a\}, \{a, b\}, \{a, b, c\}\}$$


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### Exercise 3: Membership, subset, and equality

Let:

$$A = \{a, b, c, 2, 3, 4\} \quad B = \{a, b\} \quad C = \{c, 2\} \quad E = \{a, b, \{c\}\} \quad G = \{\{a, b\}, \{c, 2\}\} \quad F = \emptyset$$

For each statement, mark it as **True** or **False**.

- a.  $c \in A$
  - b.  $c \in F$
  - c.  $c \in E$
  - d.  $\{c\} \in E$
  - e.  $\{c\} \in C$
  - f.  $B \subseteq A$
  - g.  $A \subseteq C$
  - h.  $F \subseteq A$
  - i.  $B \in G$
  - j.  $\{B\} \subseteq G$
  - k.  $G \subseteq A$
  - l.  $A = B$  iff  $(A \subseteq B \wedge B \subseteq A)$
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### Exercise 4: Union, intersection, and complement

Work in a universe (domain)  $D = \{m, r, k, f\}$ , where these are individuals, shorthand for *Mina*, *Rory*, *Kiki*, and *Fido*, respectively. Let the following one-place predicates denote sets:

$$\llbracket dog \rrbracket = \{f\} \quad \llbracket cat \rrbracket = \{k\} \quad \llbracket student \rrbracket = \{m, r\} \quad \llbracket tall \rrbracket = \{r, f\} \quad \llbracket sleepy \rrbracket = \{m, k\}$$

- a. Compute each set:

- i.  $\llbracket tall \rrbracket \cap \llbracket student \rrbracket$
- ii.  $\llbracket sleepy \rrbracket \cup \llbracket cat \rrbracket$
- iii.  $\llbracket student \rrbracket \setminus \llbracket sleepy \rrbracket$
- iv.  $D \setminus \llbracket tall \rrbracket$

b. Using your answers from (a), decide whether each statement is true or false:

- i.  $r \in \llbracket tall \rrbracket \cap \llbracket student \rrbracket$
- ii.  $k \in \llbracket sleepy \rrbracket \cup \llbracket cat \rrbracket$
- iii.  $m \in \llbracket student \rrbracket \setminus \llbracket sleepy \rrbracket$
- iv.  $f \in D \setminus \llbracket tall \rrbracket$

## Exercise 5: Intersective modification and set-theoretic meaning

Use intersective modification as set intersection:

$$\llbracket happy\ dog \rrbracket = \llbracket happy \rrbracket \cap \llbracket dog \rrbracket.$$

Work again with  $D = \{m, r, k, f\}$  and add:

$$\llbracket happy \rrbracket = \{k, f\}.$$

- a. Compute  $\llbracket happy\ dog \rrbracket$ ,  $\llbracket sleepy\ cat \rrbracket$ , and  $\llbracket tall\ student \rrbracket$ .
- b. Translate each sentence into set-theoretic form (using  $\in$  and, where helpful,  $\cap$ ):
  - i. Fido is a happy dog.
  - ii. Kiki is a sleepy cat.
  - iii. Rory is a tall student.
- c. Evaluate each sentence as true or false in this model:
  - i. Mina is a sleepy student.
  - ii. Kiki is a tall cat.
  - iii. Fido is a sleepy dog.